

Grade 6 Accelerated
Unit 6
Lesson 13 Homework

Name Answer Key
Date _____
Period _____

Evaluate each expression.

1. $(4)^5 = 1,024$

$$\begin{array}{r} 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \\ 16 \cdot 16 \cdot 4 \\ 256 \cdot 4 \end{array} \quad \begin{array}{r} 256 \\ \times 4 \\ \hline 1024 \end{array}$$

2. $(-6)^3 = -216$

$$\begin{array}{r} (-6) \cdot (-6) \cdot (-6) \\ 36 \cdot (-6) \end{array} \quad \begin{array}{r} 36 \\ \times 4 \\ \hline 216 \end{array}$$

3. $\left(-\frac{4}{5}\right)^4 = \frac{256}{625}$

$$\begin{array}{r} \left(-\frac{4}{5}\right)\left(-\frac{4}{5}\right)\left(-\frac{4}{5}\right)\left(-\frac{4}{5}\right) \\ \hline \frac{16 \cdot 16}{25 \cdot 25} \end{array}$$

4. $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$

$$\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$$

5. The expression $(-4)^4$ will result in a ☐ negative ☒ positive value because

the ☐ base ☒ exponent is an ☐ odd ☒ even number.

For questions 6-10, select all equivalent values or expressions.

6. $(2.6)^3$

- ☐ 7.8
- ☐ 17.576
- ☐ $(3)(2.6)$
- ☐ $(2.6)(6.76)$
- ☐ $2.6 \cdot 2.6 \cdot 2.6$

$$\begin{array}{r} 2.6 \\ \times 2.6 \\ \hline 156 \\ +520 \\ \hline 6.76 \end{array} \quad \begin{array}{r} 6.76 \\ \times 2.6 \\ \hline 4056 \\ +13520 \\ \hline 17.576 \end{array}$$

7. $(-4)^4$

- ☐ -20
- ☐ 256
- ☐ -256
- ☐ $(16)(16)$
- ☐ $(-4)(-16)(-4)$
- ☐ $(-4)(-4)(-4)(-4)$

$$\begin{array}{r} (-4)(-4)(-4)(-4) \\ 16 \cdot 16 \\ 256 \end{array}$$

8. $\left(-\frac{1}{3}\right)^5$

- ☐ $-\frac{1}{243}$
- ☐ $\frac{1}{243}$
- ☐ $\left(\frac{1}{3}\right)(-5)$
- ☐ $\left(-\frac{1}{27}\right)\left(\frac{1}{9}\right)$
- ☐ $\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3}$

$$\left(-\frac{1}{3}\right)\left(-\frac{1}{3}\right)\left(-\frac{1}{3}\right)\left(-\frac{1}{3}\right)\left(-\frac{1}{3}\right)$$

9. $(-5)(-5)(-5)(-5)$

- ☐ $(-5)^4$
- ☐ 625
- ☐ $(25)^2$
- ☐ -625
- ☐ $(-25)(-25)$

10. $(-7)(-7)(-7)$

- ☐ $(7)^3$
- ☐ 343
- ☐ $(-7)^3$
- ☐ -343
- ☐ $7 \cdot (-7)^2$

$$\begin{array}{r} (-7)(-7)(-7) \\ 49 \cdot (-7) \\ -343 \end{array}$$